	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 3

Lab Title: Realization of Logic gate using NAND gate

Student Name: Muammad Waqas **Reg. No:** 240756

Objective: Realization of Logic gates using universal gate as NAND GATE

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____

EXPERIMENT 03

Realization of logic gates as NAND Gate

Objectives:

- Realization of logic gates using NAND gate which is a universal gate.

Equipment required:

- TTL IC-7400
- Digital Electronics Trainer

Background Knowledge:

Universal gates are the ones which can be used for implementing any gate like AND, OR and NOT, or any combination of these basic gates. The NAND and NOR gates are universal gates. In practice, this is advantageous since NAND and NOR gates are economical and easier to fabricate and are the basic gates used in all IC digital logic families.

NAND Gate:

The NAND gate represents the complement of the AND operation. Its name is an abbreviation of NOT AND. The graphic symbol for the NAND gate consists of an AND symbol with a bubble on the output, denoting that a complement operation is performed on the output of the AND gate.

The truth table and the graphic symbol of NAND gate is shown in the figure below:

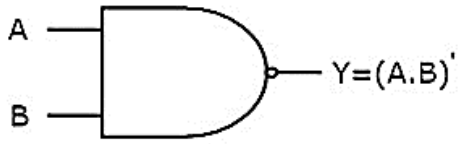


Figure 1: Logic Symbol of NAND Gate

A	B	C
0	0	1
0	1	1
1	0	1
1	1	0

Figure 2: Truth Table of NAND Gate

The truth table clearly shows that the NAND operation is the complement of the AND.

Lab Tasks:

Task 1:

Connect the circuit as shown. Complete the truth table and identify the type of operation.



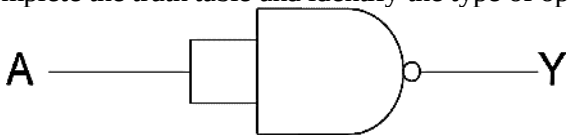
Truth Table:

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Y = **Y = ¬(A · B)** **Type of operation:** **NAND GATE**

Task 2:

Connect the circuit as shown. Complete the truth table and identify the type of operation.



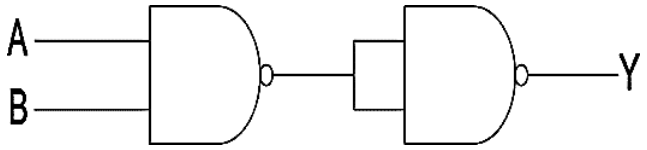
Truth Table:

A	Y
0	1
1	0

Y = **A NAND A** **Type of operation:** **NOT GATE**

Task 3:

Connect the circuit as shown. Complete the truth table and identify the type of operation.



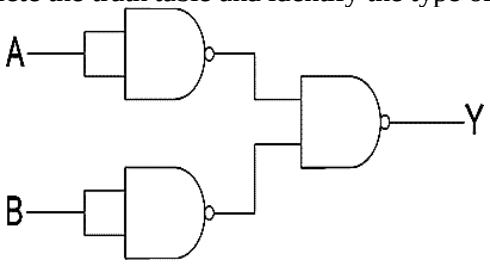
Truth Table:

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Y = (A NAND B) NAND (A NAND B) **Type of operation: AND GATE**

Task 4:

Connect the circuit as shown. Complete the truth table and identify the type of operation.



Truth Table:

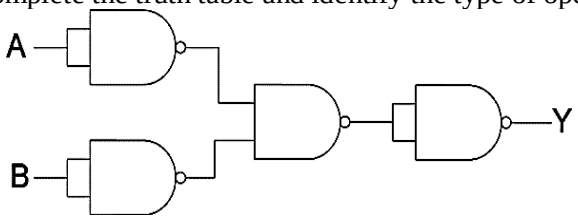
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Type of operation: OR GATE

Y = (A NAND A) NAND (B NAND B)

Task 5:

Connect the circuit as shown. Complete the truth table and identify the type of operation.



Truth Table:

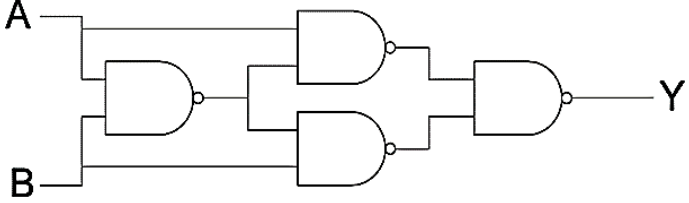
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Y = [(A NAND A) NAND (B NAND B)] NAND [(A NAND A) NAND (B NAND B)]

Type of operation: NOR GATE

Task 6:

Connect the circuit as shown. Complete the truth table and identify the type of operation.



Truth Table:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

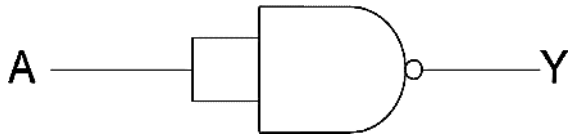
Y = A⊕B

=[(A NAND B) NAND (A NAND (A NAND B))] NAND [(B NAND (A NAND B)) NAND (B NAND (A NAN D B))]

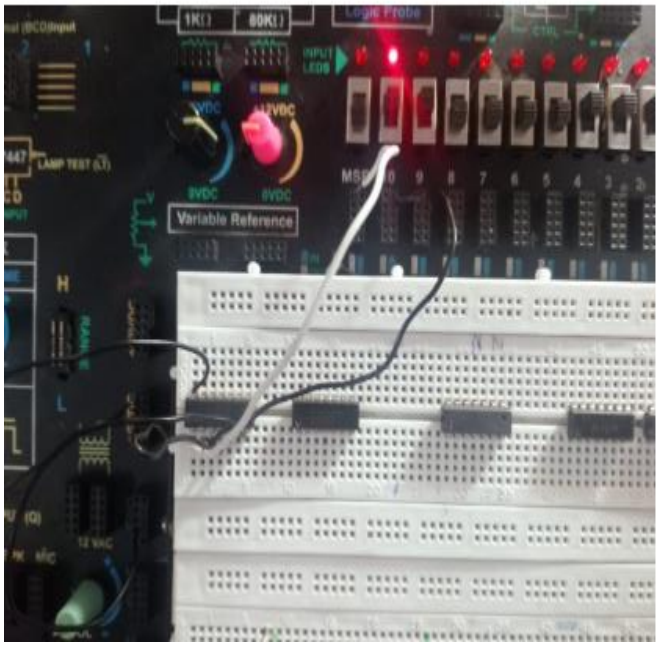
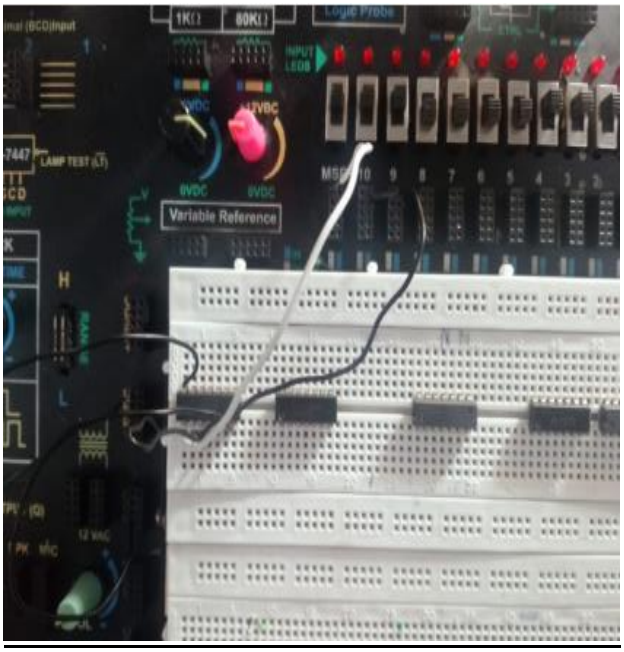
Type of operation: XOR GATE

Simulation Activities:

1. **Simulate the given circuit and identify the type of operation.**



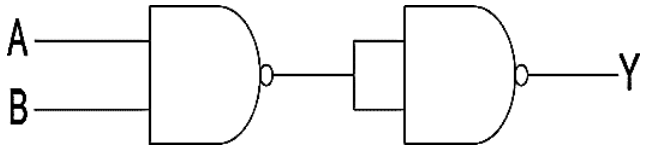
Simulation:

ON	OFF
	

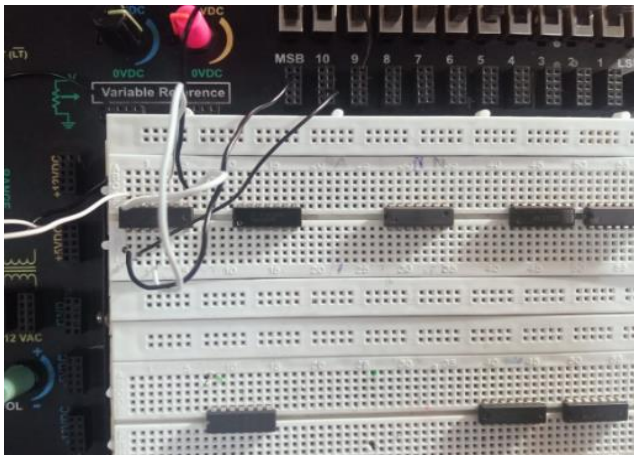
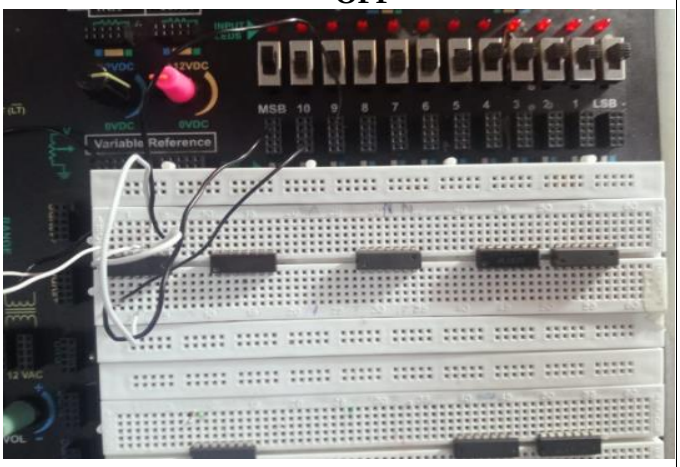
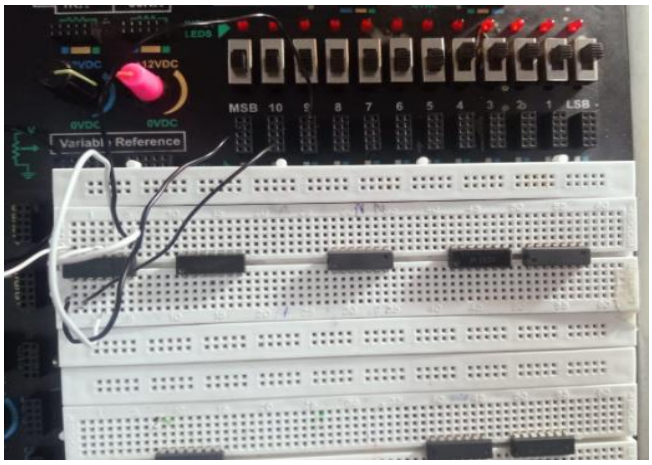
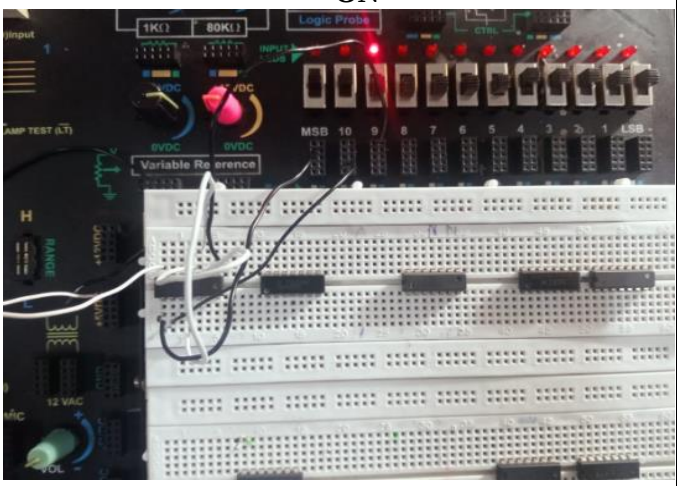
Type of Operation:

NOT GATE

2. **Simulate the given circuit and identify the type of operation.**



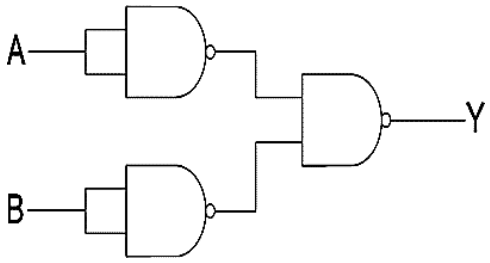
Simulation:

OFF 	**OFF** 
OFF 	**ON** 

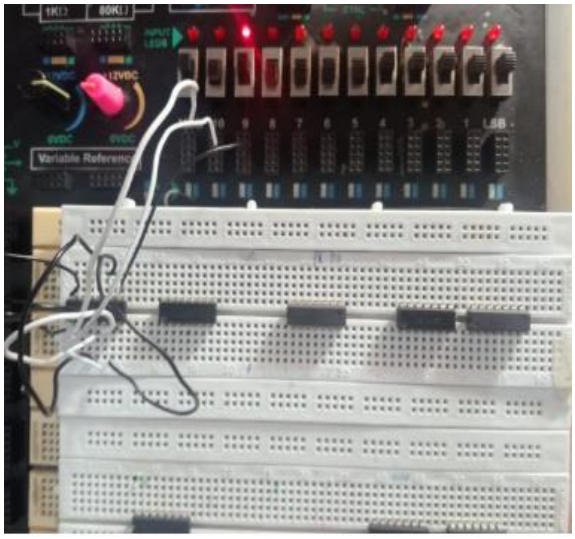
Type of Operation:

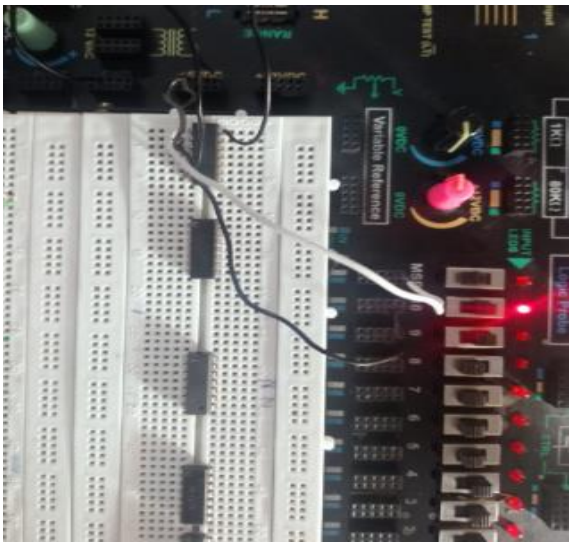
AND GATE

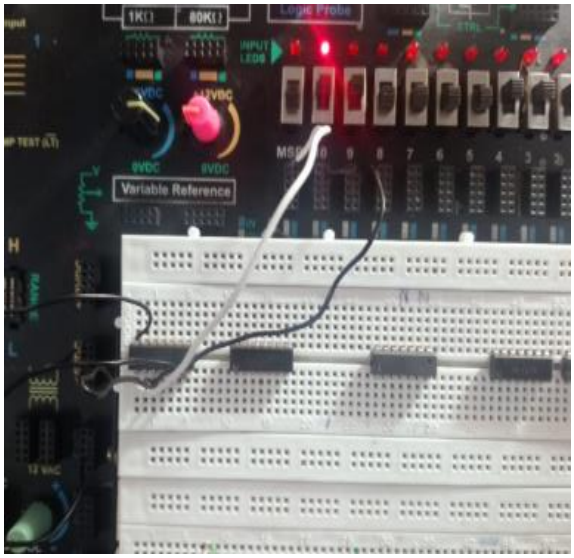
3. **Simulate the given circuit and identify the type of operation.**

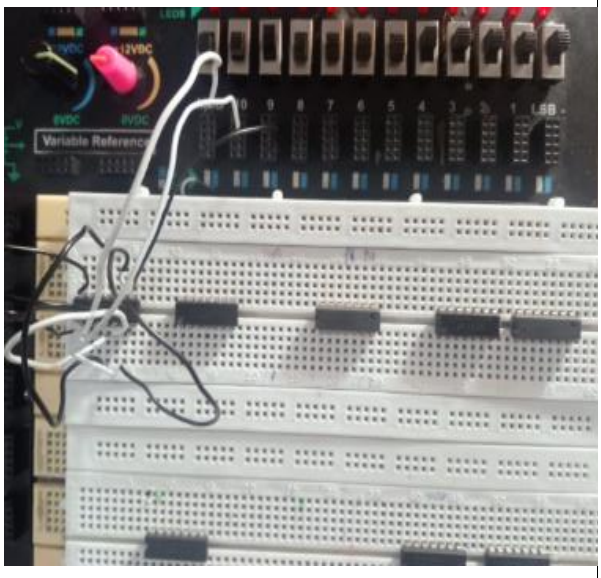


Simulation:





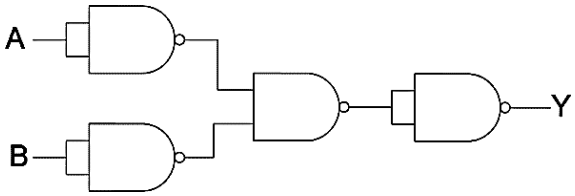




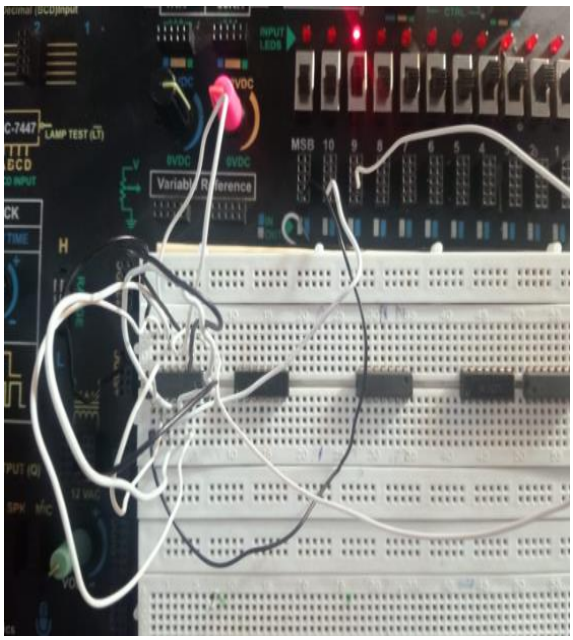
Type of Operation:

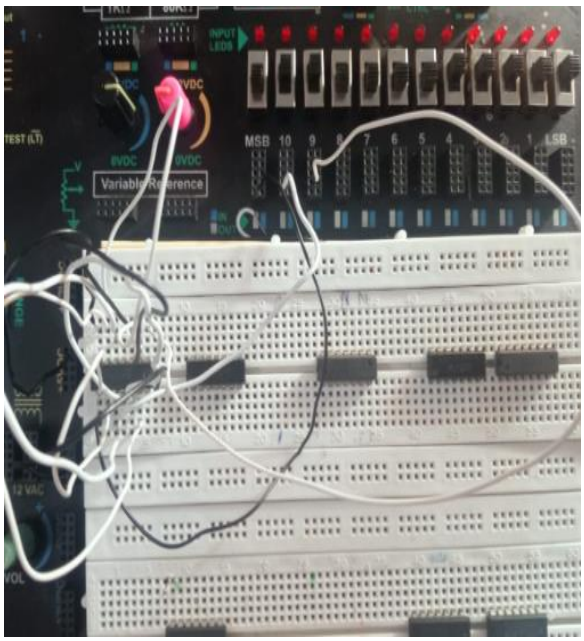
OR AGTE

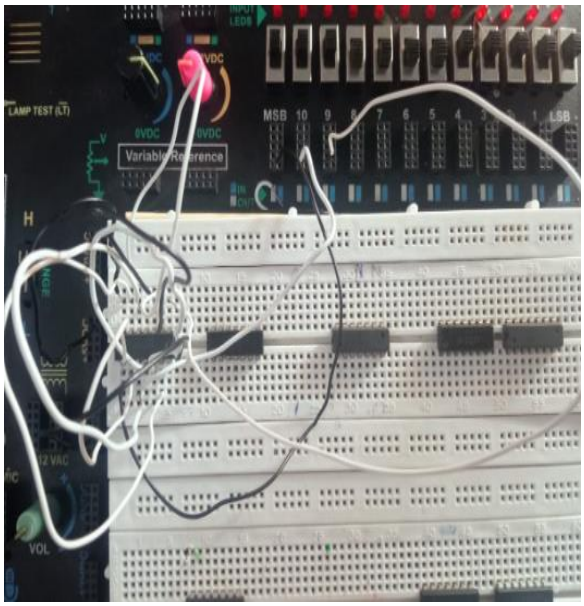
4. **Simulate the given circuit and identify the type of operation.**

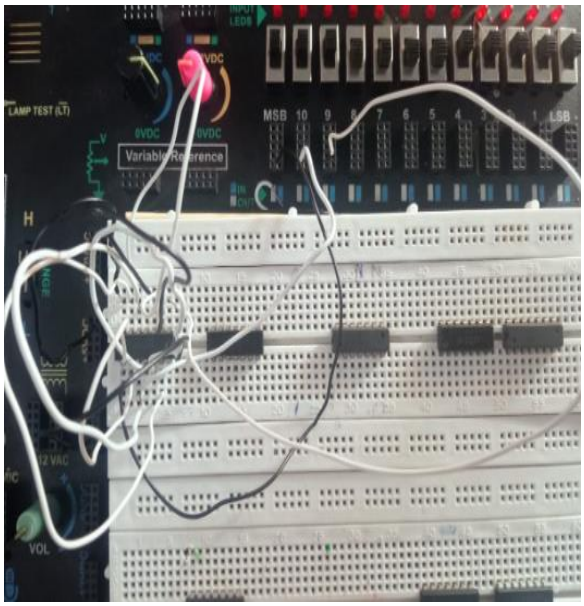


Simulation:





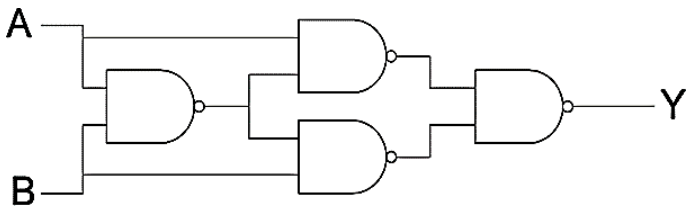




Type Of Operation

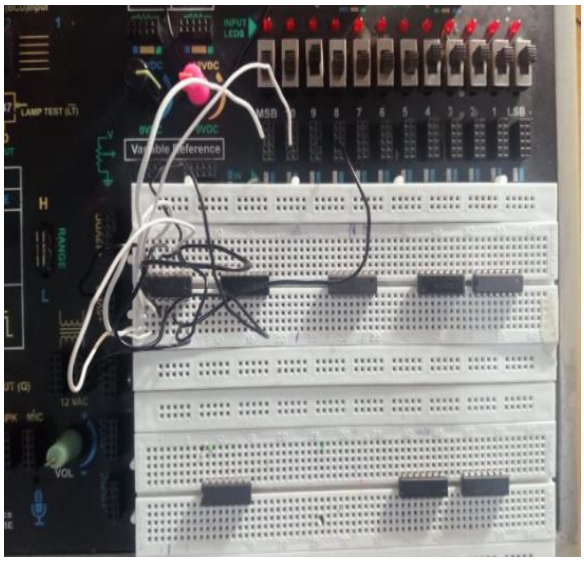
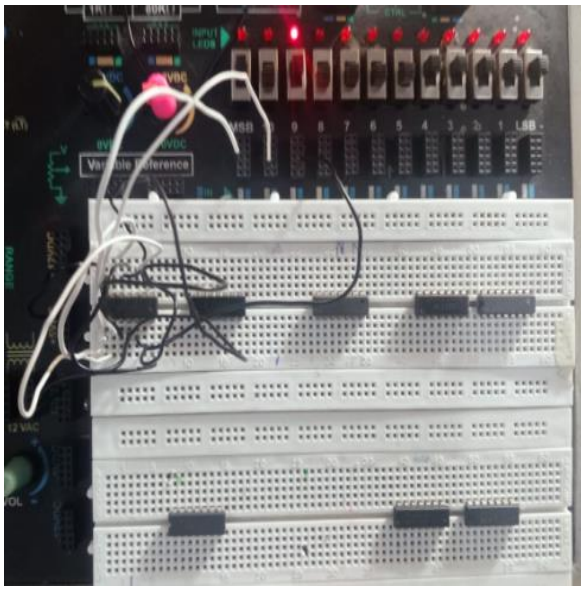
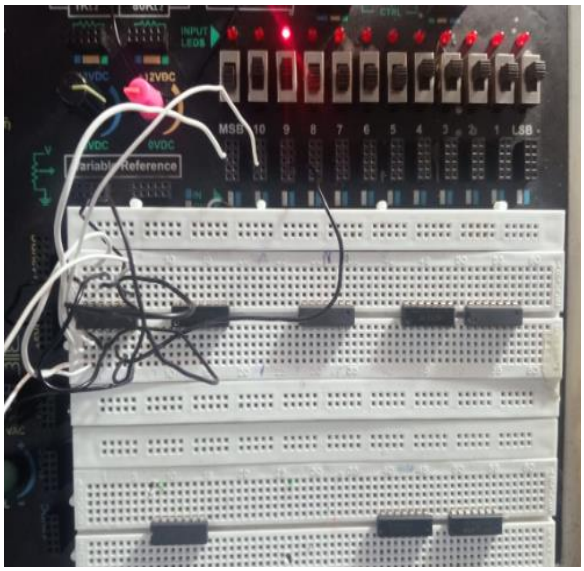
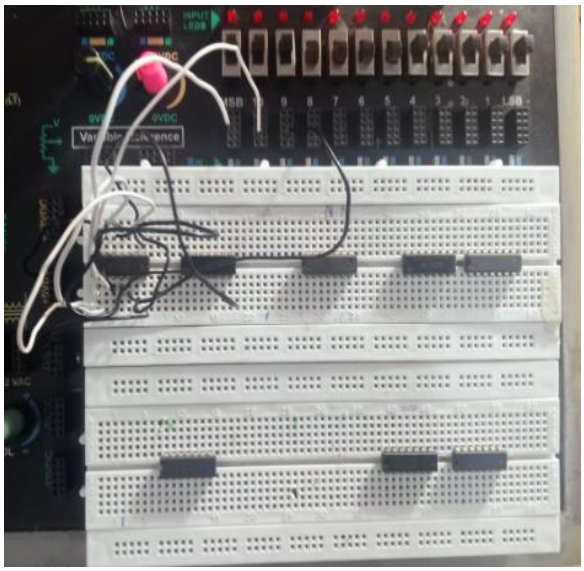
NOR GATE

5. **Simulate the given circuit and identify the type of operation.**



6.

Simulation:

OFF	ON
	
ON	**OFF**
	

Type of Operation:

XOR GATE

Conclusion:

Using the **NAND gate** as a **universal gate** is a fundamental concept in digital logic design. The circuits analyzed demonstrated that the NAND gate can be successfully configured to perform all three basic logic operations: **NOT**, **AND**, and **OR**, as well as the derived gates **NOR** and **XOR**. By simply rearranging and cascading two-input NAND gates, we effectively replicated the functionality of every other major gate. This ability confirms the NAND gate's status as a universal gate, proving that any complex logic circuit, regardless of its function, can be built using only NAND gates. This simplicity is crucial in microelectronics, allowing for manufacturing efficiency by using a single, uniform gate type to construct all digital systems.